

TK171014 - Plag Report

by Code Delieveries

Submission date: 17-Mar-2025 02:12AM (UTC-0700)

Submission ID: 2617071265

File name: Advancing_Cloud.docx (1.03M)

Word count: 5693

Character count: 32633

Advancing Cloud Data Protection with Cryptographic Algorithms

Abstract: Encryption facilitates the transit of private information through channels that are not secure and reservations regarding the unauthorized loss or modification of that data. Various encryption schemes have been developed through time to protect data in different application domains. This paper focuses on symmetric encryption, often used for key exchange due to its large key size. In particular, the AES (Advanced Encryption Standard) algorithm is widely adopted for securing sensitive data due to its efficiency and robust encryption. Additionally, the ECC (Elliptic Curve Cryptography) algorithm is explored for its lightweight yet secure approach to key exchange, offering an optimal balance between security and computational efficiency. Security holds the paramount place in data centers and cloud computing. This paper proposes a quantum computing algorithm for encrypting cloud data. This combination leverages quantum computing's rapid complex computations. The quantum computing algorithm offers fast, efficient, and secure data protection in cloud environments, reducing computational power requirements while enhancing overall efficiency. Moreover, it addresses the limitations of traditional encryption methods, including AES and ECC, by providing a scalable solution for large-scale data encryption. By integrating quantum computing, AES, and ECC, we can achieve unprecedented levels of security and performance. The potential impact of this approach could revolutionize data protection standards in the cloud industry.

Keywords: Encryption, sensitive data, insecure channels, data loss, unauthorized entities, data security, symmetric encryption, key exchange, large key size, AES, ECC, cloud computing, security, quantum computing, algorithm, rapid computations, secure data protection, computational power, efficiency, traditional encryption methods, scalable solution, large-scale data encryption, unprecedented security, data protection standards, cloud industry.

14 I. INTRODUCTION

In recent years, distributed computing has changed the manner in which organizations and people store, cycle, and access information. The cloud offers remarkable advantages, including scalability, flexibility, cost-effectiveness, and ease of access

from anywhere. However, as more sensitive information moves into cloud environments, safeguarding it against unauthorized access and potential cyber threats has become a top priority. Classic cryptographic techniques, namely encryption algorithms like RSA and AES, are primarily engaged in shielding information while electrically stored and during the process of being telegraphed. AES, as a symmetric encryption algorithm, is widely used for its speed and efficiency in securing data, but it faces challenges in key management and scalability in large cloud systems. On the other hand, ECC (Elliptic Curve Cryptography) offers a more efficient public-key encryption method, providing robust security with smaller key sizes, making it a promising solution for cloud environments where both security and computational efficiency are essential. However, these traditional methods, including AES and ECC, are facing increasing pressure to adapt to the growing scale of cloud computing infrastructures and the emerging threats posed by quantum computing. While symmetric encryption algorithms such as AES are efficient, they face challenges in terms of key management and scalability. As cloud systems expand, the amount of data being transferred and stored grows exponentially, making traditional encryption methods more cumbersome and less efficient. The extreme urgency of developing cryptographic systems that are perfectly secure and fully scalable has never been greater. Quantum computing capabilities, on the other hand, threaten to compromise the security of traditional encryption systems as the field matures further. Because of the ability of qubits to be in multiple states at once, quantum computers have the power of superior computation over classical computers. Some problems that would take several millennia on present-day classical computers could easily be solved by quantum systems in a matter of days due to such enhanced capabilities. With quantum-enabled encryption technology providing sound security protocols that resist quantum-based attacks, these hurdles may soon be circumvented. Quantum Key Distribution (QKD) is the principal application for the secure exchange of cryptographic keys with the help of the laws of quantum physics; any procedure for interception would, according to quantum mechanics, distort the quantum-encrypted data thus rendering such interception detectable. With the emergence of quantum cryptography, long-term data security is feasible against arising threats since these techniques apply quantum algorithms that resist attacks from quantum computers,

including post-quantum cryptography. On the other hand, quantum-enhanced encryption can significantly improve data encryption at a large scale. Quantum systems would allow for faster encryption techniques using quantum algorithms like Grover's Algorithm for quicker searching and Shor's Algorithm for the factorization of enormous numbers. More importantly, these systems will be less energy-consuming and resource-intensive. They serve to be specifically appropriate for cloud environments, where computing power is distributed across a number of data centers, along with high data throughput.

These quantum computing applications in the cloud environments might also allow rethinking the entire architecture of data protection systems. Instead of merely encrypting data at rest or in transit, quantum encryption could create an avenue toward new security philosophies that protect their data continuously on different levels, during processing, storage, and transmission. On the other hand, quantum encryption can offset new threats, including attacks powered by AI, which can completely undermine current encryption in almost no time. With the evolution of AI technologies in sophistication, combining quantum encryption with AI-based security systems may herald the birth of next-generation cyber-security protocols able to adapt themselves in real time to threats in evolution. At some point, in the future, this quantum computing encryption coupled with cloud should also open up an opportunity to standardize quantum-safe encryption algorithms across industries. There are some early initiatives by NIST on the standardization of post-quantum cryptographic algorithms, which will hopefully become an industry-de facto standard for cloud security. Therefore, an organization across the globe adopting encryption technologies that are robust from current and future perspectives can be guaranteed. All in all, bringing quantum computing into cloud-based encryption stands as a considerable boost toward securing sensitive data in a rapidly intertwining digital world. This transition is broad in scope considering the immaturity of quantum computing in terms of cloud security. It paves the way for developing more secure, efficient, scalable frameworks for data protection and will inevitably redefine privacy and trust in the digital age. Hence, the design and implementation of quantum encryption algorithms for cloud platforms would herald a secure and resilient digital ecosystem that can cope with quantum incorporation and extensive rapid advancements in cloud. By addressing the limitations of conventional encryption, such as AES and ECC, and harnessing the might of quantum computing, quantum-enhanced encryption can prepare cloud and enterprise environments to protect

their valuable data against any attacks in the data-centric world.

A. Objective of the study

The objectives of this research consist of creating a quantum-computing-based encryption algorithm for secure and efficient data protection in cloud environments; further addressing the limitations of conventional encryption methods in terms of scalability and computational efficiency; improving large data encryption performance; and fortifying against such changing cyber threats. In addition, the proposed study shall endeavor to cut down on computability requirements and provide a scalable encryption solution, thus setting new benchmarks for secure, efficient, and reliable data protection in cloud computing.

B. Problem Statement

Traditional encryption methods used in cloud computing face significant challenges in scalability, computational efficiency, and vulnerability to evolving cyber threats. Symmetric encryption, while effective, often struggles with large-scale data environments due to its high computational power requirements and key management complexities. As cloud computing grows, the need for secure, efficient, and scalable encryption becomes critical. This paper addresses these limitations by proposing a quantum computing-based encryption algorithm. Leveraging the speed and computational capabilities of quantum technology, this approach ensures robust data security, reduces computational demands, and offers a scalable solution for protecting sensitive data in modern cloud environments.

II. RELATED WORK

[1] In 2023, several papers were published, including a preprint on arXiv titled "Quantum Accessible Encryption for Cloud Information With Full-Blind Quantum Calculation," co-authored by Wenjie Liu, Yinsong Xu, and Wen Liu, together with Haibin Wang and Zhibin Lei.

A quantum encryption system for cloud data that executes full-blind computationally secure queries via a multi-client universal circuit is presented by the authors. Under said protocols, neighborhood quantum devices with very small numbers N of quantum bits (acceptable for limited QKD) could possibly contract with a trustworthy key generation facility and data encryption, before upload encrypted data to a data center. Incorporating Grover's search algorithm facilitates searching the encrypted data effectively with quantum security provided against all quantum rival attacks. The authors present an example of searching on an

encrypted 2-qubit state, following it up with a complete security analysis demonstrating the strength of the scheme against threats, both external and internal.

[2] In the 1st Mar 2024, the authors released a preprint titled "Towards an Original Protection Safeguarding Circulated Multiparty Information Reevaluating Plan for Distributed computing with Quantum Key Distribution" on arXiv.com. The conference was one that the authors held an investigation for.

The co-performing study explores the relationship of these cloud computing and blockchain technologies along with the quantum computing to deal with several of the symptoms of the present times. The security framework proposed by the authors enhances the protection of cloud blockchain system data by implementing The Quantum Key Distribution (QKD) is very effectively combined with the use of CRYSTALS Kyber and Zero-Knowledge Proofs (ZKPs). Relying on lattice-based cryptography, the framework tries to protect data against attacks from quantum computers in a quantum-safe fashion. According to this study, any organization that wishes to secure its data against quantum attacks will benefit from a complete system efficiency, quantum key generation rate, and encryption or decryption process.

[3] The preprint titled "PristiQ" was made available on arXiv in 2024 and describes the Co-Design Framework "PristiQ," which concerns the protection of quantum machine-learning security mechanisms in cloud environments.

The co-design framework for the first time as per the research presented herein "PristiQ" stands to address data security requirements for any QML application concerning quantum-as-a-service systems. With awareness of the risk that arises from data leakage due to running QML models on cloud-based quantum computers, the authors include a user-chosen security key with the encryption subcircuit and additional safe qubits. This ensures that data security is enhanced through quantum encryption during computation. The paper introduces another automated search algorithm that optimizes model execution of quantum data as it lies in the encrypted form. Experimental findings highlight that PristiQ offers quantum data secure protection while still ensuring the performance of QML applications. The answer consists of system tests that were carried out using IBM quantum hardware and simulation models.

[4] In 2023, Tirthak Patel, Daniel Silver, Aditya Ranjan, Harshita Gandhi, William Cutler, and Devesh Tiwari distributed a paper titled "Toward Security in Quantum Program Execution On

Untrusted Quantum Distributed computing Machines for Business-touchy Quantum Needs" as an arXiv preprint (arXiv:2307.16799).

This article shares the need for protecting privately owned and sensitive quantum code from malicious or untrustworthy agents when it comes to cloud-based quantum computing systems. To this end, the authors propose an "SPYCE" system that obfuscates quantum codes and outputs an effort to prevent leakage of sensitive data over the cloud. SPYCE serves as a low overhead, scalable, and resource-efficient solution towards enabling business-sensitive quantum computations on untrusted quantum cloud platforms as it is grounded in the unique principles of quantum computing.

[5] The article, "Quantum : The Implications of Quantum-Safe Cryptography for Your Info in the Cloud" appeared on the IBM Think Blog in 2024.

The content from IBM analyses new challenges in cyber security, especially that of data in the cloud; quantum computing has emerged as a big player in this area. As noted in the article, the extant encryption techniques designed for data in motion and at rest will be vulnerable to the attacks from large-scale quantum computers equipped with millions of fault-tolerant qubits. An immediate adoption of quantum-resistant cryptographic algorithms to combat threats of this kind is urged by IBM, that is, an urgent call to preserve against possible future attacks arising due to quantum technology.

III. PROPOSED SYSTEM

A computational model for secure quantum multicloud data with efficient motive based on quantum computing algorithm is proposed and analyzed.

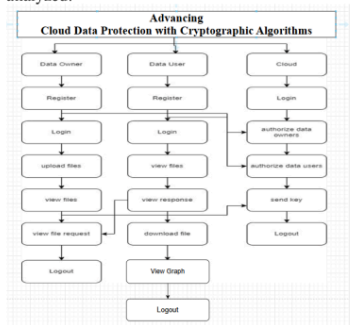


Figure 1 Flow chart of Advancing Cloud Data Protection with Cryptographic Algorithms.

Using quantum mechanics, fast pace creates encryption and decryption processes with low computational costs than classical ones. This scheme essentially resolves the problem of traditional encryptions inherent scalability through quantum-key-distribution-oriented secured key management across distributed cloud environments.

This capability integrates a strong solution for the scalable protection of large-scale data under a future-oriented approach to meet emerging threats such as quantum attacks.

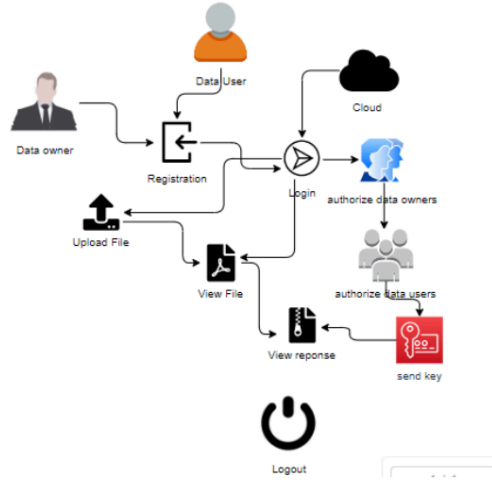


Figure 2 System Architecture of Advancing Cloud Data Protection with Cryptographic Algorithm

C. Implementation modules

We went on to apply and develop several modules in our study on Enhancing Cloud Data Security Using Cryptographic Algorithms.

Data Owner:

- Register: The data owner can register on the platform by entering all the necessary details.
- Login: The data owner can log into the platform using his credentials after obtaining the permission of the cloud administrator.
- Upload Files: A data owner can upload files on the platform and encrypt those files via quantum encryption algorithms.
- View Files: Allows the data owner to view and manage their uploaded files.
- View File Requests: This allows the data owner to check requests from data users and decrypt files when necessary.
- Logout: This allows the data owner to securely log out from the system.

Data User:

- Register: The procedure in which a data user can open an account in the platform by providing necessary information.
- Login: It allows the data user to access the platform with valid entries after the authorization by the cloud administrator.
- View Files: To allow a data user to view the files existing on the system and send requests to the data owner for access to the data.
- View Responses: To allow a data user to view the status of a file access request.
- Download Files: Allows the data user to download the files using the decryption keys.
- Logout: Allows the data user to log out from the platform in a secure manner.

Cloud Administrator:

- Login: Lets the cloud administrator access the platform with its credentials

- **Authorize Data Owners:** This feature lets administrator review and authorize or deauthorize registered data owners
- **Allow Data Users** to give the administrator of reviewable acceptance of registered data users and revoke it if necessary.
- **Send Keys:** A beneficial characteristic in the system of file security is the ability to send decryption keys privately using the user's email.
- **Logout:** Provides options to cloud administrator to safely log out from the platform

IV. METHODOLOGY

1. Quantum-Based Encryption Algorithm:

This quantum-based encryption technique exploits the special capabilities of quantum computing for the generation of cryptographic keys and carrying out encryption and decryption operations. These keys, which are generated randomly and are required to encrypt and decrypt sensitive data, are usually created by exploiting fundamental quantum behaviors like the Hadamard gate, which creates superposition states. Thus, quantum randomness is combined with classical methods of cryptography to give a secure and effective way of safeguarding data.

Quantum Key Generation:

The central feature of this algorithm is the creation of a random cryptographic key using a quantum circuit. The create_random_key function generates this key by applying Hadamard gates on each qubit in the quantum circuit. The Hadamard gate is a key operation in quantum computing that puts the qubit in a superposition of states, resulting in a more randomized output.

After applying the Hadamard gates, the circuit measures the states of the qubits. These measurements yield a random binary string, which becomes the cryptographic key. Quantum randomness ensures that the key is unpredictable and difficult for any adversary to replicate without access to the quantum system itself.

Text to Binary Conversion:

The Text_to_binary function is converting plaintext that needs to be encrypted to a binary representation. This binary representation is required for the XOR-based encryption-and-decryption transfer process. Each character in the text is converted into its ASCII value, which is then represented as an 8-bit binary string.

Encryption:

The encrypt_text function takes a bitwise XOR (exclusive OR) operation between the binary representation of the text and the key generated by quantum means. Theoretically, XOR encryption can

be treated as a simple but efficient way of scrambling the data. In the present case, the binary text is encrypted by going from the least significant to the most significant bit, XORing each bit with the corresponding bit in the key. The key will wrap around and be repeated if it is shorter than the text. This creates an encrypted binary string, i.e., ciphertext.

Decryption:

The decrypt_text function "undoes" encryption by means of an XOR operation. An XOR operation is symmetrical in that the same key that encrypts input can decrypt it. The original binary text is recovered when two encrypted binary strings are XORed with the same key and passed to a binary_to_text function to recover the original.

Advantages of Quantum Key Generation

Quantum Randomness: The generation of Keys using quantum mechanics is where this algorithm gets its strongest point. Classic random number generators (RNG), according to an algorithm, can be predicted in an excellent manner if the attacker has sufficient computational power. In the realm of quantum randomness, however, it is purely random and cannot be predicted. Hence, this makes quantum-generated keys much more secure, being protected from any suspicious generation with near-perfect randomness. Each key produced by a quantum system then appears to be unique and non-oppositional.

Security: XOR is a simple technique, but when applied with quantum-generated keys, it can provide robust security. With its large keyspace, brute force is nearly impossible, particularly when keys are randomly generated through a quantum system. The randomness of quantum mechanics itself greatly decreases the likelihood that the key may be inferred or reconstructed by the intruder.

Efficiency: Quantum gates such as the Hadamard gate allow for relatively fast key generation, and being computationally light, the XOR encryption scheme can go hand in hand with applications where efficiency is a key factor. The encryption and decryption processes are typically very fast, even for the larger data sets.

3CC Algorithm for Data Encryption:

Elliptic Curve Cryptography is a kind of public key cryptography that uses the master's mathematics of elliptic curves over finite fields to provide encryption, digital signatures, and key exchange methods. ECC seeks to provide the same security offered by conventional public-key algorithms, such as RSA, but with smaller key sizes, thereby making possible greater efficiency and usability in modern computing environments.

The fundamental mechanism for ECC is based on elliptic curves, which are smooth and continuous curves through a descriptive equation. These curves share properties having become necessary for the successful execution of cryptographical operations. Typically, the elliptic curve discrete logarithm problem must be extremely difficult to solve. As such, the security of ECC resides in the difficulty of this problem, leading the algorithm to be highly resistant against a wide array of attacks from brute-force attacks to factorization attacks, which are known to work against other public-key algorithms such as RSA.

Key Generation:

Elliptic Curve Cryptography (ECC), a key pair cryptographic algorithm, employs the theory of elliptic curves over fields that are necessarily algebraically closed. In general, First, choose a base point P on the elliptic curve to create an ECC key pair that, in later calculations, will remain unchanged. The private key is generally a random integer (called d), whereas the corresponding public key is produced by multiplying the base point P with the private key d . The product of this operation is a point on the curve, named $Q=dP$; this is the public key. ECC uses much smaller keys than traditional cryptographic algorithms like RSA, but ECC gives equivalent or even more security than RSA, thus making it very efficient.

Encryption Process:

The primary operations of ECC encryption involve using algorithms such as ECDH or Elliptic Curve Diffie-Hellman key exchange for securely exchanging keys, or it can use Elliptic Curve ElGamal for the service. In the scheme of ECC-based encryption, the sender and the receiver are agreed upon a common elliptic curve and also a base point. Following this procedure, Alice generates a private key, d_A , and pairs it with a public key, $Q_A=d_AP$. The receiver has a private key d_B , public key $Q_B=d_BP$. In the encryption process, a sender generates a shared secret using the receiver's public key and the sender's private key, which is then applied to derive symmetric encryption keys for the secure encryption of the message. The strength of ECC encryption comes from the difficulty of solving the elliptic curve discrete logarithm problem (ECDLP). It is difficult to infer the private key from the public key for standard protocols.

Decryption Process:

Decryption in ECC is the reverse of encryption. The receiver, having the private key d_B and the encrypted message, can use their private key along with the sender's public key to calculate the shared secret key. Once the shared secret is derived, it can be used to decrypt the ciphertext. The decryption process relies on the same elliptic curve arithmetic

used for encryption. The key challenge in ECC decryption is solving the elliptic curve discrete logarithm problem, which is computationally difficult even for powerful adversaries, thus ensuring the confidentiality of the data.

Advantages of ECC:

ECC is an efficiency and scalability. The prime one is the comparability regarding key size small compared to RSA equivalent security. For example, a key of 256 bits in ECC brings the same security as RSA with 3072 bits, which means faster encryption, decryption, and key exchange. Less cumulative overhead, reduced need for bandwidth, less storage requirements favor built ECC systems in limited resources like mobile and IoT devices and cloud platforms. Smaller key sizes require less power consumption, and hence it matters in battery-operated devices, making ECC appropriate for modern high-performance applications such as safe web traffic, mobile encryption, and VPN.

Security:

ECDLP renders ECC as robust cryptography. This problem is computationally infeasible to solve using classical computing methods, making ECC resistant to traditional cryptographic attacks such as brute force and factorization. The security of ECC increases with key size, and it is considered highly secure even with smaller key sizes compared to other algorithms like RSA. However, like any cryptographic system, ECC's security depends on the careful selection of elliptic curves and key management practices. Additionally, with the potential rise of quantum computing, ECC may eventually face challenges from quantum algorithms like Shor's algorithm, which could efficiently solve the elliptic curve discrete logarithm problem. As a result, post-quantum cryptographic methods are being researched to ensure long-term security.

Advanced Encryption Standard:

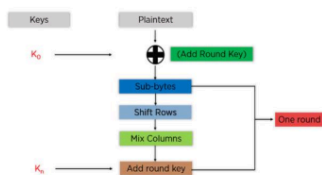
The AES name stands for Advanced Encryption Standard; it is also an encryption algorithm of a symmetric type. It is the encryption standard chosen by the U.S. government, and there is almost universal agreement that AES is among the very best encryption in terms of security. On data blocks, the AES process works in a standard size of 128 bits. The corresponding lengths of various sections of this process key are usually 128, 192, or 256 bits in accordance with security levels required. AES is a running series of rapid encrypt/decrypt processes based on a pre-determined number of rounds depending on the size.

- **128-bit key:** 10 rounds of encryption
- **192-bit key:** 12 rounds of encryption
- **256-bit key:** 14 rounds of encryption

AES functions by executing data encryption through a series of steps, each of which entails the manipulation of a 4x4 matrix representing an individual data block. Within this matrix, each cell contains one byte of data. Considering that a block comprises 16 bytes, the matrix comprehensively encapsulates the entirety of the block's data.

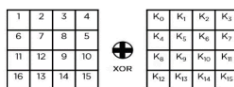
0	1	2	3
4	5	6	7
8	9	10	11
12	13	14	15

The state array, illustrated in the accompanying diagram, plays a pivotal role in the AES (Advanced Encryption Standard) algorithm. In addition to that, the primary key will be expanded into a total of $(n + 1)$ Plant ont ves towards AES encryption, which is specifying their number n for the rounds in the encryption. For instance, in case of using 128 bit Key and for 16 rounds then maximum of 11 keys will be generated (i.e. $10 + 1$). AES Encryption Stages consist of processes like SubBytes, ShiftRows, MixColumns, and finally Apply AddRoundKey.iteration state appropriate for the number of rounds. The aforementioned discussed steps change the input plaintext into ciphertext using the keys generated from them. This article briefly introduces the constituents of the AES algorithm as well as the steps through which it processes encryption.



This process necessarily follows an orderly set of operations for each block and then combines them into the address for one ultimate ciphertext. Stages would follow in the order: Adding a round key: This amounts to a XOR operation on the block data in the stated array and the first generated key (K0). This is

followed by feeding the output state array to the next



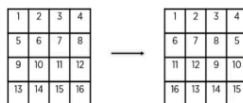
operation.

State-by-state State-substitution on each byte indicates the process whereby a substitution is applied to each byte in the state array with every byte being changed to 2 segments indicating rows and columns in hex format, replaced by new values from an S-Box-by-picture or contesting state array.



Shift Rows is an operation of data modification entailing the interchange of row elements. It usually leaves the first row intact and shifts the elements of the other rows left by a certain number of places. For example, In a four-row matrix, the two-row elements can shift an amount of one point on the left. The three-row elements could shift an amount of two points on the left. Likewise, the elements in the last row can shift by three left points. This kind of operation is an integral part of the security enhancement and ease of distribution of several encryption methodologies, including the Advanced Encryption Standard (AES).

Shift Rows is really an operation pertaining to data manipulation in which they can interchange row elements. Normally, it keeps the first row static and moves other's components in subsequent rows after a certain number of places to the left. For example, elements present in row two of four-row matrix would be shifted one place to the left; while those present in the third row would displace two positions left. Likewise, the final row's elements could shift three positions to the left. Such an operation, important for security improvement and ease of distribution in several encryption methods including the advanced encryption standard (AES), is the basis of some encryption methods.



The mixing columns stage is correspondence in the AES encryption method. This stage multiplies all the columns in the state array by a fixed matrix to create an entirely new column for the next state array. This process continues for all the columns of the state array into state arrays for the next step. This is to be underscored as it is performed in all rounds of encryption, except in the last round.

$$\begin{array}{|c|c|c|c|} \hline 1 & 2 & 3 & 4 \\ \hline 5 & 6 & 7 & 8 \\ \hline 9 & 10 & 11 & 12 \\ \hline 13 & 14 & 15 & 16 \\ \hline \end{array} \times \begin{array}{c} C_0 \\ C_1 \\ C_2 \\ C_3 \end{array} = \begin{array}{c} NC_0 \\ NC_1 \\ NC_2 \\ NC_3 \end{array}$$

Constant Matrix Old Column New Column

From the SubBytes operation followed by ShiftRows, and MixColumns of the data of the first round, the round key is then added. The round function begins with XORing the array of states with the key for that round. If the array of states is from the last round, it will output the block's ciphertext. Otherwise, it serves as the new input state array for the subsequent round.

$$\begin{array}{|c|c|c|c|} \hline 1 & 2 & 3 & 4 \\ \hline 6 & 7 & 8 & 5 \\ \hline 11 & 12 & 9 & 10 \\ \hline 16 & 13 & 14 & 15 \\ \hline \end{array} \oplus \begin{array}{|c|c|c|c|} \hline K_0 & K_1 & K_2 & K_3 \\ \hline K_4 & K_5 & K_6 & K_7 \\ \hline K_8 & K_9 & K_{10} & K_{11} \\ \hline K_{12} & K_{13} & K_{14} & K_{15} \\ \hline \end{array}$$

XOR

The extracted State array must be passed on as input to the subsequent round, observing the similar operating of the same as heretofore discussed. Here is what you should do for that:

Round Key Addition:

$$\begin{array}{|c|c|c|c|} \hline 6F & 8F & 4F & 2F \\ \hline 7F & 0F & 0F & 1F \\ \hline 0F & 0F & 0F & 0F \\ \hline 0F & 0F & 0F & 0F \\ \hline \end{array} \oplus \begin{array}{|c|c|c|c|} \hline 00 & 3C & 63 & A7 \\ \hline 9F & 4E & 29 & 36 \\ \hline 7E & 48 & 16 & 31 \\ \hline 54 & 55 & 08 & 1A \\ \hline \end{array}$$

Plaintext Round 0 Key New State Array

Sub-Bytes: To obtain a whole new state array, the elements are passed through a 16x16 S-Box.

63	EB	9F	A0
CO	2F	93	92
AB	30	AF	C7
20	CB	2B	A2

New State Array

Shift Rows:

63	EB	9F	A0
CO	2F	93	92
AB	30	AF	C7
20	CB	2B	A2

Old State Array

63	EB	9F	A0
2F	93	92	CO
AF	C7	AB	30
A2	20	CB	2B

New State Array

Mix Columns:

$$\begin{array}{|c|c|c|c|} \hline 02 & 03 & 01 & 0F \\ \hline 0E & 04 & 05 & 04 \\ \hline 05 & 01 & 02 & 05 \\ \hline 03 & 01 & 01 & 02 \\ \hline \end{array} \times \begin{array}{|c|c|c|c|} \hline 63 & EB & 9F & A0 \\ \hline 2F & 93 & 92 & CO \\ \hline AF & C7 & AB & 30 \\ \hline A2 & 20 & CB & 2B \\ \hline \end{array} = \begin{array}{|c|c|c|c|} \hline BA & 64 & EB & B0 \\ \hline 75 & A4 & BD & 40 \\ \hline F4 & BD & 06 & 7D \\ \hline 7A & 32 & 0E & 5D \\ \hline \end{array}$$

Constant Matrix Old State Array New State Array

Add Round Key:

$$\begin{array}{|c|c|c|c|} \hline BA & 64 & EB & B0 \\ \hline 75 & A4 & BD & 40 \\ \hline F4 & BD & 06 & 7D \\ \hline 7A & 32 & 0E & 5D \\ \hline \end{array} \oplus \begin{array}{|c|c|c|c|} \hline E2 & 01 & 07 & 06 \\ \hline 32 & 02 & 05 & 29 \\ \hline FC & 9B & E4 & A2 \\ \hline F1 & BB & E4 & 93 \\ \hline \end{array}$$

Old State Array Round 1 Key New State Array

The past round-final ciphertext is now that state array. For this round, this state array serves as input to round number two. Repeat steps one through ten until round 10 is achieved; at that point, you'll have the final cipher text, subject to length of the key.

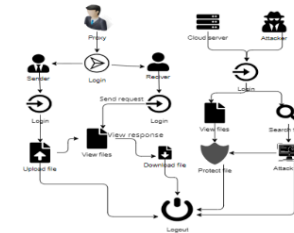
29	57	40	1A
C3	14	22	02
50	20	99	D7
5F	F6	B3	3A

Final State Array after Round 10

AES Final Output

29 C3 50 5F 57 14 20 F6 40 22 99 B3 1A 02 D7 3A

Ciphertext



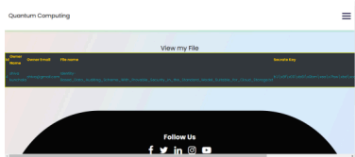
V. RESULTS AND DISCUSSION

These are the results of Architecture of Advancing Cloud Data Protection with Cryptographic Algorithm

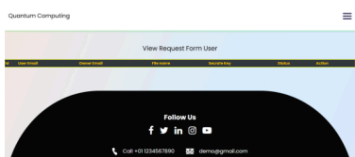
Uploading Files: Here data owner can upload files into website.



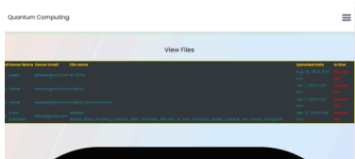
View Uploaded Files: Here data owner can view the uploaded files.



View Requests from Users: Here data owner can view the file requests of users.



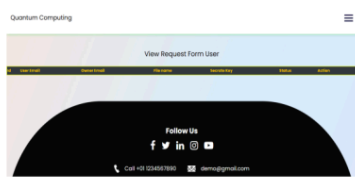
View Files: Here data user can view the uploaded files and they are able to request files.



View Responses: Here data user can view the Request Response and he can able to download the file.



Sending Key to the User: Here cloud can send the keys to requested users .



VI. CONCLUSION AND FUTURE WORK

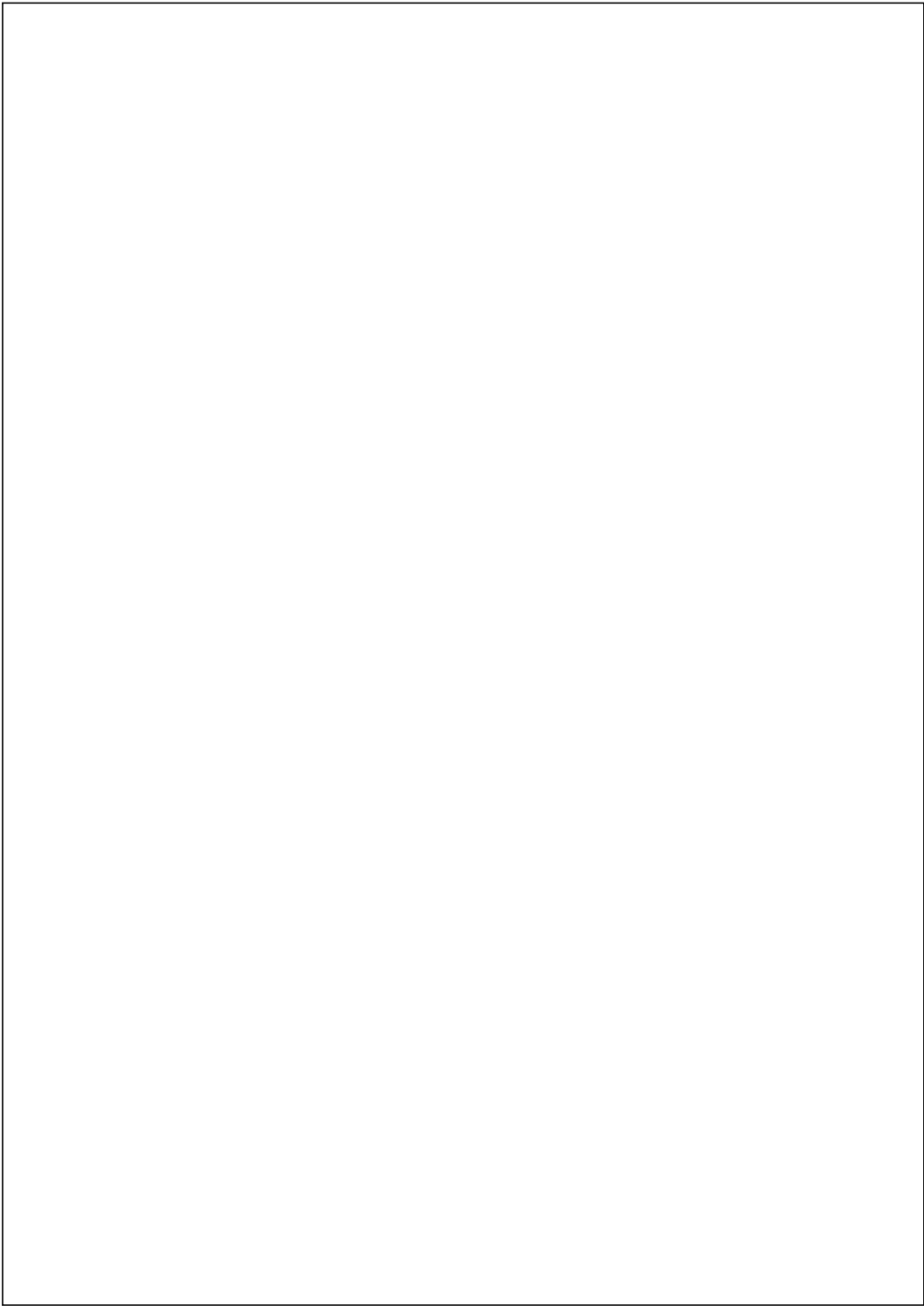
The integration of quantum computing algorithms into cloud data encryption represents a significant advancement in cybersecurity. By leveraging principles such as quantum key distribution and quantum-safe cryptography, this approach addresses the vulnerabilities inherent in classical encryption methods, particularly against emerging quantum threats. Implementing these quantum algorithms enhances data protection, ensuring that sensitive information remains secure even as quantum technologies evolve. As quantum computing continues to develop, adopting quantum-resistant encryption standards becomes imperative for organizations aiming to safeguard their data against future risks. This proactive strategy not only fortifies current security measures but also future-proofs data protection frameworks, maintaining the confidentiality and integrity of information in an increasingly complex digital landscape.

As quantum computing progresses, it will be vital to further enhance cloud data encryption against newly created threats. This future work will incorporate quantum-resistant algorithms, such as lattice and code-based cryptography, to protect the data from future quantum attacks. The implementation of Quantum Key Distribution (QKD) will strengthen further security in key exchanges by detecting an attempt at eavesdropping according to the principles of quantum mechanics. Coupled with improvements in homomorphic encryption, this will permit computations to be performed on data while still encrypted, thus ensuring confidentiality during processing. Constant research addressing these algorithms' optimization for speed and scalability will, therefore, ensure top-notch protection of sensitive details stored in cloud environments, keeping data integrity and privacy in a quantum space.

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